Different Types of Graphs and Charts for Presenting Data

To better understand each chart and graph type and how you can use them, here's an overview of graph and chart types.

Bar Graph

A bar graph should be used to avoid clutter when one data label is long or if you have more than 10 items to compare.

<u>Best Use Cases for These Types of Graphs:</u>

Bar graphs can help you compare data between different groups or to track changes over time. Bar graphs are most useful when there are big changes or to show how one group compares against other groups.

The example above compares the number of customers by business role. It makes it easy to see that there is more than twice the number of customers per role for individual contributors than any other group.

A bar graph also makes it easy to see which group of data is highest or most common.

For example, at the start of the pandemic, online businesses saw a big jump in traffic. So, if you want to look at monthly traffic for an online business, a bar graph would make it easy to see that jump.

<u>Other use cases for bar graphs include:</u>

Product comparisons
Product usage
Category comparisons
Marketing traffic by month or year
Marketing conversions

<u>Design Best Practices for Bar Graphs:</u>

Use consistent colors throughout the chart, selecting accent colors to highlight meaningful data points or changes over time.
Use horizontal labels to improve readability.
Start the y-axis at 0 to appropriately reflect the values in your graph.

<u>When to use bar charts</u>

- If you have more than 10 items or categories to compare.
- If your category labels or names are long.

- Comparing parts of a bigger set of data, highlighting different categories, or showing change over time.
- Have long categories label — it offers more space.
- If you want to illustrate both positive and negative values in the dataset.
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Best practices for bar charts

- Focus on one color for a bar chart. Accent colors are ideal if you want to highlight a significant data point.
- Bars should be wider than the white space between bars.
- Write labels horizontally (not vertically) for better readability in your bar chart.

When to avoid:

- If you're using multiple data points.
- If you have many categories, avoid overloading your graph. Your graph shouldn't have more than 10 bars.
- Order categories alphabetically or by value to ensure consistency across your bar chart.

Column Chart

Use a column chart to show a comparison among different items, or to show a comparison of items over time. You could use this format to see the revenue per landing page or customers by close date.

Best Use Cases for This Type of Chart:

While column charts show information vertically, and bar graphs show data horizontally. While you can use both to display changes in data, column charts are best for negative data.

For example, warehouses often track the number of accidents that happen on the shop floor. When the number of incidents falls below the monthly average, a column chart can make that change easier to see in a presentation.

In the example above, this column chart measures the number of customers by close date. Column charts make it easy to see data changes over a period of time. This means that they have many use cases, including:

Customer survey data, like showing how many customers prefer a specific product or how much a customer uses a product each day.

Sales volume, like showing which services are the top sellers each month or the number of sales per week.

Profit and loss, showing where business investments are growing or falling.

Design Best Practices for Column Charts:

Use consistent colors throughout the chart, selecting accent colors to highlight meaningful data points or changes over time.

Use horizontal labels to improve readability.

Start the y-axis at 0 to appropriately reflect the values in your graph.

Line Graph

A line graph reveals trends or progress over time and you can use it to show many different categories of data. You should use it when you chart a continuous data set.

Best Use Cases for These Types of Graphs:

Line graphs help users track changes over short and long periods of time. Because of this, these types of graphs are good for seeing small changes.

Line graphs can help you compare changes for more than one group over the same period. They're also helpful for measuring how different groups relate to each other.

A business might use this type of graph to compare sales rates for different products or services over time.

These charts are also helpful for measuring service channel performance. For example, a line graph that tracks how many chats or emails your team responds to per month.

Design Best Practices for Line Graphs:

Use solid lines only.
Don't plot more than four lines to avoid visual distractions.
Use the right height so the lines take up roughly 2/3 of the y-axis' height.

When to use line charts

- Compare and present lots of data at once.
- Show trends or progress over time.

- Highlight deceleration.
- Present forecast data and share uncertainty in a single line chart.
- If you have a continuous dataset that changes over time.
- If your dataset is too big for a bar chart.
- If you want to display multiple series for the same timeline.
- If you want to visualize trends instead of exact values
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Best practices for line charts

- Use solid lines only because dotted or dashed lines are distracting.
- Ensure that points are ordered consistently.
- Label lines directly and avoid using legends in a line chart.
- Don't chart more than four lines to avoid visual distractions.
- Zero baseline is not required, but it is recommended for a line chart.

When to avoid:

1. Line charts work better with bigger datasets, so, if you have a small one, use a bar chart instead

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Pie Chart

A pie chart shows a static number and how categories represent part of a whole — the composition of something. A pie chart represents numbers in percentages, and the total sum of all segments needs to equal 100%.

Best Use Cases for This Type of Chart:

The image above shows another example of customers by role in the company.

The bar graph example shows you that there are more individual contributors than any other role. But this pie chart makes it clear that they make up over 50% of customer roles.

Pie charts make it easy to see a section in relation to the whole, so they are good for showing:

Customer personas in relation to all customers
Revenue from your most popular products or product types in relation to all product sales
Percent of total profit from different store locations

Design Best Practices for Pie Charts:

Don't illustrate too many categories to ensure differentiation between slices.
Ensure that the slice values add up to 100%.
Order slices according to their size.

When to use pie charts

- Illustrate part-to-whole comparisons — from business to classroom charts and graphs.
- Identify the smallest and largest items within data sets.
- Compare differences between multiple data points in a pie chart.
- When you show relative proportions and percentages of a whole dataset.
- Best used with small datasets — also applies to donut charts.
- When comparing the effect of ONE factor on different categories.
- If you have up to 6 categories.
- When your data is nomial and not ordinal.
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Best practices for using a pie chart

- Limit categories to 3-5 to ensure differentiation with the pie chart slices.
- Double-check if the total value of the slices is equal to 100 percent.
- Group similar slices together in one bigger slice to reduce clutter.
- Make your most important slice stand out with color. Use shades of that specific color to highlight the rest of the slices.
- Order slices thoughtfully. For example, you can place the largest section at the 12 o'clock position and go clockwise from there. Or place the second largest section at the 12 o'clock position and go counterclockwise from there.

When to avoid:

- If you have a big dataset.
- If you want to make a precise or absolute comparison between values.

Scatter Plot Chart

A scatter plot or scattergram chart will show the relationship between two different variables or reveals distribution trends. Use this chart when there are many different data points, and you want to highlight similarities in the data set. This is useful when looking for outliers or for understanding the distribution of your data.

Best Use Cases for These Types of Charts:

Scatter plots are helpful in situations where you have too much data to quickly see a pattern. They are best when you use them to show relationships between two large data sets.

In the example above, this chart shows how customer happiness relates to the time it takes for them to get a response.

Great use cases for this type of graph make it easy to see the comparison of two data sets. This might include:

Employment and manufacturing output
Retail sales and inflation
Visitor numbers and outdoor temperature
Sales growth and tax laws

Try to choose two data sets that already have a positive or negative relationship. That said, this type of graph can also make it easier to see data that falls outside of normal patterns.

Design Best Practices for Scatter Plots:

Include more variables, like different sizes, to incorporate more data.
Start the y-axis at 0 to represent data accurately.
If you use trend lines, only use a maximum of two to make your plot easy to understand.

When to use a scatter plot

- Show relationships between two variables.
- To show correlation and clustering in big datasets.
- If your dataset contains points that have a pair of values.
- If the order of points in the dataset is not essential.
- You have two variables of data that complement each other.

Best practices for scatter plots

- Start the y-axis value at zero to represent data accurately.
- Plot additional data variables by changing dot sizes and colors.
- Highlight with color and annotations.

When to avoid:

1. If you have a small dataset.
2. If the values in your dataset are not correlated.

Heat Map

A heat map shows the relationship between two items and provides rating information, such as high to low or poor to excellent. This chart displays the rating information using varying colors or saturation.

Best Use Cases for Heat Maps:

In the example above, the darker the shade of green shows where the majority of people agree.

With enough data, heat maps can make a viewpoint that might seem subjective more concrete. This makes it easier for a business to act on customer sentiment.

There are many uses for these types of charts and graphs. In fact, many tech companies use heat map tools to gauge user experience for apps, online tools, and website design.

Another common use for heat map graphs is location assessment. If you're trying to find the right location for your new store, these maps can give you an idea of what the area is like in ways that a visit can't communicate.

Heat maps can also help with spotting patterns, so they're good for analyzing trends that change quickly, like ad conversions. They can also help with:

Competitor research
Customer sentiment
Sales outreach
Campaign impact
Customer demographics

Design Best Practices for Heat Map:

Use a basic and clear map outline to avoid distracting from the data.
Use a single color in varying shades to show changes in data.
Avoid using multiple patterns.